XR CHARACTER DIRECTION / INTERACTION LANDSCAPE / 2023–2026

What has to happen between the actor and the target?

Compare scene directions involving another generated actor, a creature, or a prop. Each annotation states what the study produces.

◇ Director instruction ◆ Demonstrated control

RT Real-time result reported

NO No real-time control result reported

Y What does the interaction require?

Response to a cue or state change

“When B starts speaking, turn toward them.”

NO **MoLaM · ICCV 2025**
Generate one character’s reaction from the other character’s supplied motion clip. Output is clip-based.

“When the handler beckons, have the dog come over.”

“When the box reaches the mark, lower it onto the table.”

RT **CLoSD · ICLR 2025**
Detect when the pelvis reaches the sofa's sitting area, then advance the task to standing.

NO **MaskedMimic · ACM TOG 2024**
Switch from approach to sit when within range of the supplied chair. Real-time throughput is unreported for this example.

Coupled motion move together over time

“Dance together. Stay in step through the turn.”

NO **Multi-Character Retargeting · SIGGRAPH 2023**
Retarget a reference salsa dance to simulated characters using interaction landmarks to preserve relative motion.

NO **InterGen · IJCV 2024**
Generate two people jumping in the same rhythm from one text prompt.

“Play tug with the dog. Let it pull the rope toward itself.”

NO **Move-in-2D · CVPR 2025**
Generate a rider’s motion in a horse-riding scene. The generated output is the rider’s motion.

“Lift the box, carry it to the table, and set it down.”

NO **CHOIS · ECCV 2024**
Lift, carry and set down a box using text, initial human and object states, and object waypoints (Fig. 3).

NO **CG-HOI · CVPR 2024**
Generate the carrying motion for a large box from its supplied trajectory (Fig. 1).

NO **Physical Character-Scene Interactions · SIGGRAPH 2023**
Simulate lifting and carrying a box to a goal using a task-specific policy.

Contact touch or support

“Give B a high-five with your right hand.”

NO **InterGen · IJCV 2024**
Generate a two-person embrace from a text prompt.

NO **InterControl · NeurIPS 2024**
Make two dancers hold hands at selected intervals while avoiding torso collisions (Fig. 4).

“Stroke the dog's back with your left hand.”

NO **Move-in-2D · CVPR 2025**
Generate the actor’s petting motion in a scene containing a dog. This is actor-motion evidence.

“Put both hands on the box handles.”

NO **Physical Character-Scene Interactions · SIGGRAPH 2023**
Use a supplied object bounding box to approach and sit on a chair in simulation.

Spatial relation without contact

“Stand beside the other actor and face them.”

NO **InterGen · IJCV 2024**
Generate a two-person bow from text. The partners face each other without a contact task.

“Stand beside the dog and look down at it.”

“Walk up to the box and stop beside it.”

NO **Motion in 3D Scenes · CVPR 2024**
Ground the table near a sofa in a 3D scan and generate a walk to it (Fig. 1).

Actor–actor
two in-scene characters

Actor–creature
actor with a nonhuman character

Actor–prop
object or furniture

X What does the actor interact with?

The creature examples from Move-in-2D establish actor motion. Joint actor–creature motion and cue-driven creature response remain open in the selected evidence.

The cue row includes clip-conditioned reactions and automatic controllers. Paper titles link to the sources.

XR VIRTUAL PRODUCTION / EQUIPMENT LANDSCAPE / 2023–2026

How do filmmakers specify equipment control?

Lighting, lens and composition, and rig motion share the same axes. Each study label identifies digital output or physical execution.

◇ Director instruction ◆ Demonstrated control

RT Real-time result reported

NO No real-time control result reported

Y What sets the timing?

Runtime condition — cue or scene feedback

“On my clap, dim the key to 30%.” / “Match the LEDs to the changing sunset feed.”

RT **Synchronized Real–Virtual Lighting · Drexel thesis 2025**
Physical lights. Map changing Unreal Engine pixel values to Aputure fixtures via wireless DMX.

“Keep her on the left third as she crosses.”

RT **CineMPC · IEEE T-RO 2024**
Physical drone and lens. Adjust flight, zoom and focus to meet image-placement and distance goals (E6).

RT **Periscope · CHI EA 2023**
Physical arm. Track the worker’s hand and adjust the camera to keep it in view.

[Show a reference frame] “Dolly toward this view and avoid anything in the way.”

RT **IRIS cinema robot arm · CRV 2026**
Physical arm. Use a supplied goal image and live feedback to push in toward a cup while avoiding an obstacle.

[Trace] “Stay on this route and keep her centered as she crosses.”

Action order — or recorded sync

“Fade up the key, then pan left.”

“Frame the piano, then pan to the bicycle.”

NO **ChatCam · NeurIPS 2024**
Virtual camera. Pan along a piano, then turn toward a bicycle from a text instruction (Fig. 4).

[Set two poses] “Move from here to here, then hold.”

NO **IRIS cinema robot arm · CRV 2026**
Physical arm. Execute a supplied waypoint sequence to measure repeatability.

[Point] “Frame here first, then reveal the doorway.”

Duration, rate — or timestamp

“Fade the key to 30% over two seconds.”

RT **CineMPC · IEEE T-RO 2024**
Simulated lens and flight. Ramp focal length from 35 to 450 mm for the dolly zoom (E3).

“Make the dolly zoom slower.”

[Trace] “Follow this arc in four seconds.”

[Trace] “Follow this route in four seconds and keep her centered.”

NO **TKCAM · NeurIPS 2026 accepted**
Digital camera. Generate a path from text and RGB keyframes supplied at selected times.

No explicit timing — single setting or goal

“Set the key to 30%.” / “Set the lens to 50 mm.”

RT **AR Lighting Adjustment · Book chapter 2025**
Physical light. Set a selected fixture’s dimming or color temperature via an AR proxy and DMX.

RT **Relightable Gaussian Splatting for VP · arXiv 2026**
Digital relighting. Change background light after capture. Renderer: about 35 fps.

“Give her a warmer key.” / “Frame the doorway behind her.”

NO **ChatCam · NeurIPS 2024**
Virtual composition. Frame the Opera House with the Harbour Bridge behind it (Fig. 1).

[Point] “Aim the camera here.” / [Trace] “Follow this dolly path.”

RT **Periscope · CHI EA 2023**
Physical arm: click a look-at point or drag to orbit.

NO **GEN3C · CVPR 2025**
Digital: rerender a clip along a supplied camera path.

RT **StageCraft Oscar · SIGGRAPH Talk 2025**
Physical LED wall: position Lightcards via tablet.

[Point] “Keep her on this third and leave the doorway in view.”

Direct settings	**Desired shot or look**	**Explicit spatial guide**	**Goal + spatial guide**
parameter values or signals	described outcome or framing goal	point, path, pose or image	outcome paired with a guide

X What does the filmmaker supply?

Runtime conditions include cues and automatic feedback. A filled mark groups related evidence. Each label describes the actual demonstration.

CineMPC’s dolly zoom is simulated. Its physical tests use motorized focus and zoom with fixed aperture. Study titles link to sources.